

Fundamentals of Meteorological Observations

Unit 3: Automatic Weather Stations & Automatic Rain Gauges

Trainer notes — for internal training use on the Capacity Connect platform.

Learning Objectives

- Explain how AWS and ARG units extend IMD's observation coverage
- Describe typical AWS sensor suites and data transmission methods
- Identify common AWS maintenance and data-gap issues

Why AWS/ARG Matter

India's terrain and population density make staffing a conventional observatory at every location impractical. AWS and ARG units close this gap by logging temperature, humidity, pressure, wind, and rainfall automatically and transmitting the data at short intervals (commonly every 15 minutes to an hour) over GPRS/satellite links back to central servers, dramatically densifying the observation network, particularly for rainfall monitoring used in flood forecasting.

Typical Sensor Suite & Power

A standard AWS carries a temperature/humidity probe, a barometric sensor, a tipping-bucket or weighing rain gauge, and an ultrasonic or cup-and-vane wind sensor, all connected to a data logger. Most rural AWS units run on solar power with battery backup, since grid power is unreliable in many installation locations - power failure is one of the most common causes of data gaps.

Maintenance & Data Quality

Because AWS units are often unstaffed, routine field visits are needed to clean solar panels, check for insect or bird interference in tipping buckets, verify sensor calibration, and confirm communication links are active. Data from AWS still passes through the same quality-control checks as manual observations before being accepted into IMD's operational and climate archives.

Summary

AWS and ARG networks are what make near-real-time, fine-grained rainfall and weather monitoring possible across India, but they only deliver reliable data when field maintenance keeps pace with the network's growth.